Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Execute the Coursera Lab Jupyter notebook for this lesson, which implements the AEKF.

1 / 1 point

Compared with the EKF of Lesson 3.1.7, which estimator produces lower root-mean-squared errors (RMSE) and which produces more reliable confidence bounds?

- ( ) The EKF produces lower RMSE and more reliable confidence bounds.
- (•) The EKF produces lower RMSE and the AEKF produces more reliable confidence bounds.
- ( ) The AEKF produces lower RMSE and more reliable confidence bounds.
- ( ) The AEKF produces lower RMSE and the EKF produces more reliable confidence bounds.

Correct
Yes, this is correct.

2. For the same Coursera Lab Jupyter notebook, compare the final AEKF estimates of $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ to the true values. (You can find the true values in the `model` variable.) Which of the following is true? For this question, consider an estimate to be "good" if it is within a few percent of the true value.

1 / 1 point

- ( ) The AEKF estimates of both $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ are good.
- ( ) The AEKF estimate of $\Sigma_{\tilde{w}}$ is good but the AEKF estimate of $\Sigma_{\tilde{v}}$ is not good.
- (•) The AEKF estimate of $\Sigma_{\tilde{w}}$ is not good but the AEKF estimate of $\Sigma_{\tilde{v}}$ is good.
- ( ) The AEKF estimates of both $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ are not good.

Correct
Yes. This is correct. But, even so, the state estimates and bounds are reasonable.

3. For the same Coursera Lab Jupyter notebook, write down the RMS state estimation error and percentage of time that the error is outside of bounds. Then, change the variables `NV` and `NW` both to 3000. Rerun the code. Which of the following is true? (If you need to change the code back to the original configuration, note that the default values for `NV` and `NW` are both equal to 1000.)

1 / 1 point

- ( ) Both the RMS state-estimation error and the percentage of time that the error is outside of bounds are better than before making the change.
- (•) The RMS state estimation error is better than before and the percentage of time that the error is outside of bounds is worse than before making the change.
- ( ) Both the RMS state-estimation error and the percentage of time that the error is outside of bounds are worse than before making the change.
- ( ) The RMS state estimation error is worse than before and the percentage of time that the error is outside of bounds is better than before making the change.

Correct
Yes, this is correct.